

<https://bkbellingham.com/>

Keith Bellingham

Portland, Oregon

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Introduction

As a multi-disciplinary physical scientist, I work with a variety of technologies, analyze data, and drive sales. I specialize in the business and development of agricultural and environmental data collection platforms. Willing to work remote with 30% travel, valid driver's license. Graduate education in physical science, 20+ years' experience. More information about my projects and publications can be found at bkbellingham.com.

Work Experience

Multi-disciplinary Physical Scientist / Business Development

Stevens Water Monitoring Systems, Inc. Feb 2006 – Present
Portland, Oregon

Stevens Water is a technology manufacturer of scientific instrumentation, environmental sensors, data loggers and cloud-based data acquisition solutions. As a physical scientist, I have a very multi-disciplinary role that includes sales, business development, and product and solution development. I have a range of technical duties that include data analyses, statical modeling to support sales efforts. My duties include:

Business

- Sales, Business development, and price negotiations.
- International distribution and sales channel management.
- Create market messaging content for key markets such as governments, academia, and agriculture.
- Provide training and tech support to both distributors and end users.
- Work with quality systems, CRMs and MRP systems.
- Manage and generate sales of electromagnetic soil sensors, and other remote deployable sensor technologies for environmental monitoring applications.
- Attending and speaking at conferences, visiting customers, submitting papers to scientific journals and trade publications, and networking.

Technical

- Data analyses using python, and Excel on sensors data for product development and technology evaluation.
- Research and development of electromagnetic soil sensor technology. Work with a team of engineers to make improvements to existing technologies and develop new technologies.
- Supervised model training using linear and polynomial regressions. Validate models using statistics for sensor calibrations and performance.
- Time series data analysis, forecasting and modeling.
- Research applications for soil moisture sensors in areas such as irrigation, agronomy, draught forecasting, and water resources to explore new markets.
- Develop and promote irrigation scheduling systems that include soil sensors, telemetry, and cloud-based software for irrigators, agronomists, and horticulture.
- Interpret hydrological data for customers using sensors and data acquisition systems.
- Install environmental monitor systems in the field.

Soil Scientist/Hydrologist

US Department of Agriculture NRCS, Internship, Feb 2003 - Dec 2004
Hillsboro, Oregon

My role at the USDA NRCS was part of a program to improve fish habitat on irrigated agricultural land. I worked with stream flow and soil data sets to assess agricultural effects on streams.

- Developed water quality monitoring program for watersheds on agricultural land to improve salmonid habitat. The water quality monitoring program statistically identifies stream reaches that fail to meet regulatory criteria.
- Constructed mathematical models that correlated water quality and stream flow on agricultural lands.
- Used statistical methods on stream flow data, GIS and soil data sets to understand water movement.

Chemical Engineer

DuBois Chemicals, Inc. 1998 - 2001
Cincinnati, Ohio

DuBois Chemical is a manufacturer of industrial chemicals specializing in surfactants. As a chemical engineer, I developed formulations of industrial fluids.

- Performed research, product development, and troubleshooting for industrial applications of metalworking fluids and other industrial chemical products.
 - Developed chemical formulations of metalworking fluids, and corrosion inhibitors with surfactants and other chemical components for industrial applications.
 - Develop chemical synthesis with starting materials to simulate large scale production.
 - Assessed environmental impacts of industrial fluid formulations through chemical and microbiological tests such as COD/BOD.
 - Provide tech support for sales professionals and end users.
 - Worked with customers and sales representatives to champion company's products at a technical level.
 - Performed chemical tests with laboratory instrumentation including liquid and gas chromatography, chemical identification spectroscopy methods, electrochemical tests, and other physicochemical methods.
 - Created and used statistical models on product performance data for engineering improvements and troubleshooting.
- Worked with ISO and other quality systems

Skills

- Sales Channel Development, price negotiations, with scientific instruments in international distribution, research, and government markets.
- Communicate the meaning of data to customers and stakeholders.
- Data Science, Machine Learning, Python, Data analyses
- Time series data modeling, forecasting trends and modeling water transport.
- Subject matter expert in soil science, environmental science, vadose zone hydrology, ground and surface water hydrology.

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- Hydrological flow models for soil water, streams, and groundwater.
- Statistics, mathematics. Normal distributions, curve fitting mathematical functions, and interpreting raw data. Work with data bases.
- Customer data bases, SQL to make queries in CRMs, and MRP systems.
- Technical writing, peer reviewed technical articles, magazine articles, manuals, and procedures. years)
- Tech support and troubleshooting scientific instrumentation.
- Irrigation Scheduling for row crops and indoor horticulture.

Education

MS in Environmental Science and Engineering

Oregon Graduate Institute at OHSU School of Science and Engineering - Portland, OR
2001 to 2003

BS in Chemistry

University of Cincinnati – Cincinnati, OH
1991 to 1996

Professional Certifications

Certified Professional Hydrologist (PH)

December 2012 to December 2019

Certification for water flow in streams, groundwater, and soil. <https://www.aihydrology.org/>
American Institute of Hydrology. #12-HWQ-4022

Certified Irrigation Auditor

June 2006 to June 2009

Irrigation Association <https://www.irrigation.org/> Certification Number ID #65086

Organizations

Soil Science Society of America
American Geophysical Union
European Geoscience Union

Publications

Bellingham, B. K. 2013. "Ch16, Sensor Based Irrigation Scheduling,." In *Evapotranspiration: Principles and Applications for Water Management*, by Megh R. Goyal, 221-231 and the Forward. Oakville, ON Canada: Apple Academic Press, Inc.

Bellingham, B. K. 2013. "New Quality Assurance Technique for Meteorological Stations Based on a Physically-Based Hydrological Model." *Western Snow Conference* 79-90.

Bellingham, B. K. January 2009. "Physicochemical Parameters of Natural Waters." *Environmental Expert.com*.

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Bellingham, B. K. 2018. Protecting Met Stations from Lightning. Application Note, Portland, Oregon: Stevens Water Monitoring Systems, Inc. .

Bellingham, B. K. 2011. "Proximal Soil Sensing, Book Review." *Vadose Zone Journal* 10:1-2 (doi:10.2136/vzj2011.0105br).

Bellingham, B. K. 2015. Sensor Based Irrigation Scheduling in Blueberries. Vol. 3, in *Sustainable Micro Irrigation Management for Trees and Vines*, by Megh R. Goyal, edited by Megh R. Goyal, 181-191. Oakville, Ontario: CRC Press, Apple Academic Press, Taylor & Francis Group.

Bellingham, B. K. 2019. "Soil Geomorphology, A Pedological Guide for Soil Sensor Applications." Tech Note, Publish for The American Geophysical Fall Meeting (Stevens Water Monitoring Systems, Inc.).

Bellingham, B. K. September 2017. "Soil Moisture Monitoring, Down to Earth." *Meteorological Technology International* 126-127.

Bellingham, B. K. September 2018. "Soil Moisture Sensors, Flow Analysis." *Meteorological Technology International* 148-149.

Bellingham, B. K. 2010. "Wireless real-time soil data used in contaminated soil cleanup operation." *Environmental Science and Engineering Magazine*.

Bellingham, B. K., J. Lea. July 16-18, 2014. "Improved Water Supply Forecasts Using Soil Moisture Data." Fourth International Symposium on Soil Water Measurements Using Capacitance, Impedance and Time Domain,. Montreal, Quebec, Canada.

Hayes, E. T., B. K. Bellingham, H. B. Mark, Jr., and A. Galal. 1996. "An amperometric aqueous ethanol sensor based on the electrocatalytic oxidation at a cobalt-nickel oxide electrode." *Electrochimica Acta* 41 (2): 337-344.

Summary

More than 20 years of professional experience in science, engineering, and business. Graduate education in the physical sciences. Manage and generate sales channels and distribution. Over a dozen technical publications. Knowledge, skills and abilities to champion a variety of technologies across, industrial, research and governmental markets.